

Electric motor

Electric motor: It is a rotating device, which rotates or moves round and round. It converts electrical energy to mechanical energy.

Working principle of electric motor: Left hand Rule.

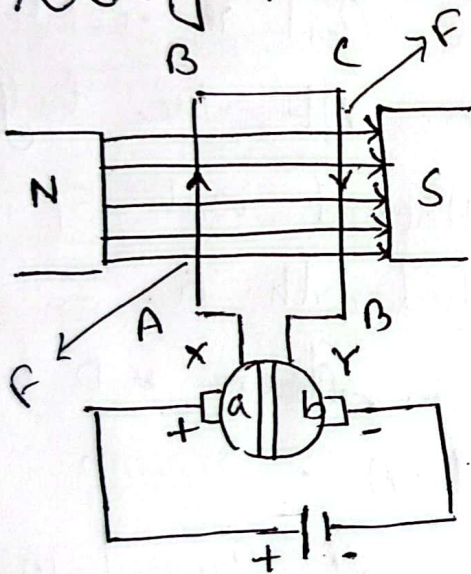


fig-1

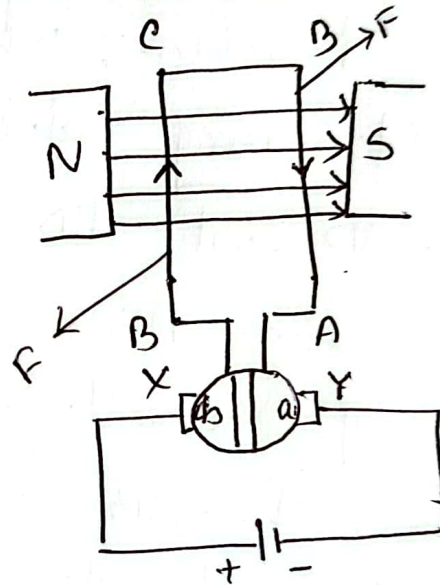


fig: 2.

When the ~~Switch~~ Battery is Switched ON, Current flows through coil from A to B. and magnetic field from N to S. So, by Fleming's left hand rule, downward force is produced on AB coil. upward force is produced in C to D coil. (Current flows from

(e to f) the Coil rotates with CD moves upward and AB moves downward. Split ring a is connected with AB and CD is connected with b. Split ring. After half rotation

AB and CD interchange their position. To change the direction of Current we use a Commutator. A Commutator consists of split rings & brushes. When the coil rotates, the split ring also rotates. At the half rotation the coil CD is attached with split ring b and AB coil attached with a. So direction of current in CD is ~~from~~ D to C and coil AB is B to A. means reverses the direction of ~~force~~ of current in the coil applying Fleming's left hand rule on CD coil the direction force is downward and the direction of force AB coil is upward.

Means, the coil rotates continuously we

This reversing of electric current happens in every half rotation and the coil continues to rotate until the battery is switched off.

Generation

An ~~electric~~ electric generator is a device which converts mechanical energy to electrical energy.

[illegible]

DC Generation: DC electric generation Constant

Q. 1. A Rectangular coil of wire ABCD.

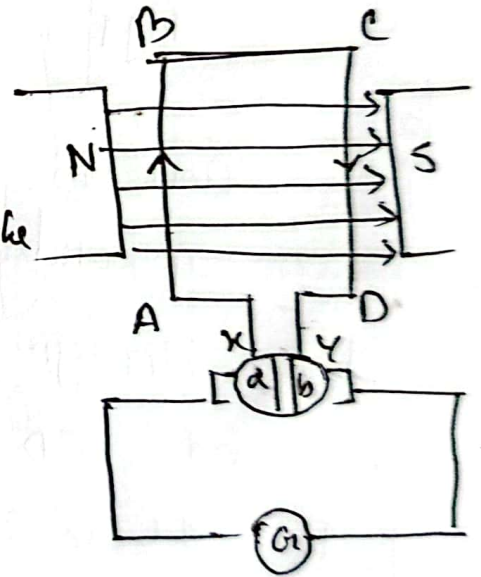
2) A Strong Magnet

3) The coil is perpendicular to the magnet

4) The end coils are connected to split rings. a and b

5) ~~to~~ Br. Rings are Connected to the branches x, y .

2) Brushes are attached to a galvanometer to show the flow of the current in the circuit.



Working of Generator:

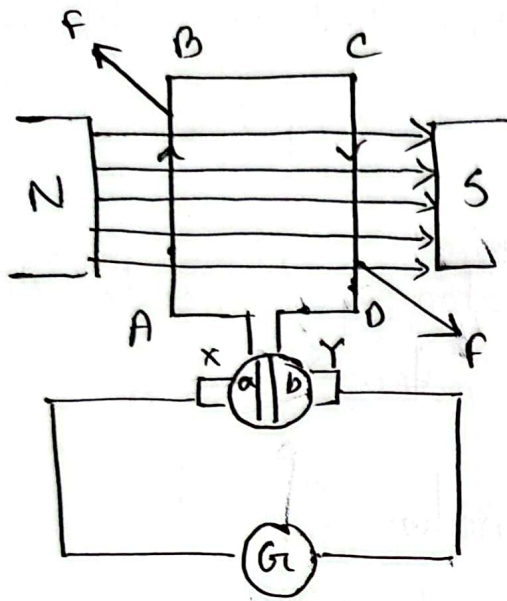


Fig 1

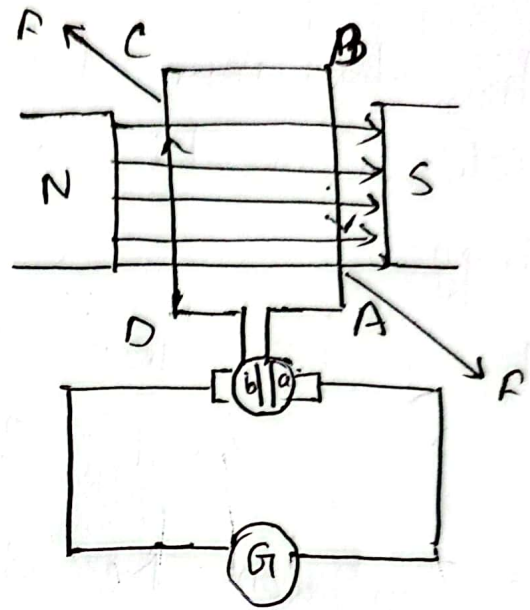


Fig 2

to
here coil is rotate clockwise. AB of coil moves upwards and CD moves downwards.
At p Applying Fleming right hand rule on AB and CD. AB is upwards and Magnetic field lines in left to right. Current flows from A to B. And CD is downward and magnetic field in left to right. Current flows from C to D.

Current flows into brush 'Y' moves along galvanometer and enters 'X'. And current flows from 'Y' to 'X' in the circuit (Fig 1)

After half rotation, CD ~~is~~ and AB interchange their position.

Split ring b is connected with CD and a is connected with ~~to~~ AB. which keeps the the direction of current in circuit same. Due to commutation the direction of current is unchanged.

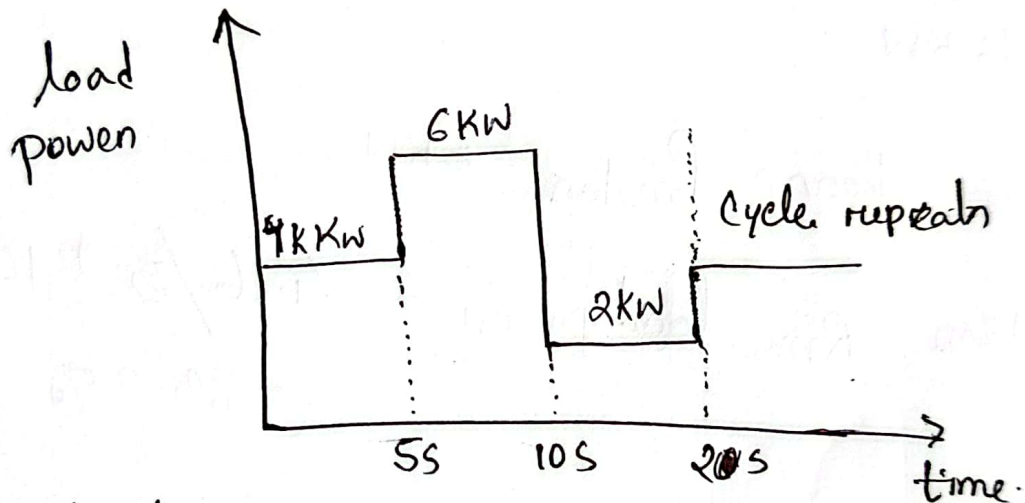
Hence the current flows from brush Y and enter X. ~~Current is same~~ Current flows from Y to X.

Thus after every half rotation, direction of current is same, so DC current is produced.

Motor Sizing: It refers to the process of picking the correct motor for a given load. It is important to size a motor correctly because:

- If a motor is too small for an application. It will not be able to control the load, leading to overshoot and ringing. If it does get the load up to the speed, the motor will overheat & burn.
- If a motor is too large for an application, then money has been wasted in purchasing large motor. Also motor typically operate inefficiently when they are run well below rated load. So, it costs money loss.

The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive the load assuming a service factor of 20%. you don't need to consider starting performance.



Peak load power requirement = 6 kW

Peak load power must be less than Rated

motor power $\times$ Service factor $\rightarrow$ Service

$$\Rightarrow 6 \text{ kW} \leq P_{\text{motor}} \times 1.2$$

$$\Rightarrow P_{\text{motor}} \times 1.2 \geq 6 \text{ kW}$$

$$\Rightarrow P_{\text{motor}} \geq \frac{6}{1.2}$$

$$\Rightarrow P_{\text{motor}} \geq 5 \text{ kW}$$

$$P_{\text{motor}} = 5 \text{ kW}$$

$$\begin{aligned}
 \text{factor} &= 1 + \frac{\text{Service factor}}{100} \\
 &= 1 + \frac{20}{100} \\
 &= 1.2
 \end{aligned}$$

$$\text{RMS Load power} = \sqrt{\frac{(P_1^2 \times t_1) + (P_2^2 \times t_2) + (P_3^2 \times t_3) + \dots + P_n^2 t_n}{t_1 + t_2 + t_3 + \dots + t_n}}$$

$$= \sqrt{\frac{(4^2 \times 5) + (6^2 \times 5) + 2^2 \times 10}{5 + 5 + 10}}$$

400

$$5 - 0 = 5$$

$$10 - 5 = 5$$

$$20 - 10 = 10$$

$$= 4.16 \text{ KW.}$$

If we choose $P_{\text{motor}} = 5 \text{ KW.}$

$$\text{then Rms load power} = \frac{4.16}{5} \times 100\% = 83.2\%$$

✓

Spring - 2025

Service factor 30%

Peak load power = 30 kW = $1 + 30/100\%$ Rated motor power
 $= 1.3$ ✓

Peak load $\geq$ Service factor $\times$ p. motor

$$P_{\text{motor}} \times 1.3 \leq \text{Peak } 30 \text{ kW}$$

$$P_{\text{motor}} \leq 30/1.3$$

$$P_{\text{motor}} = 23.07 \text{ kW}$$

$$\text{Rms load power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \dots + P_n^2 t_n}{t_1 + t_2 + t_3 + \dots + t_n}}$$

$$= \sqrt{\frac{10^2 \times 9 + 16^2 \times 4 + 20^2 \times 5 + 30^2 \times 2 + 24^2 \times 4}{9 + 4 + 5 + 2 + 4}}$$

$$= \sqrt{\frac{900 + 1024 + 2000 + 1800 + 2304}{24}}$$

$$= \sqrt{334.5}$$

Visomox®

= 18.28 kW Refresh Tears®

Zymarini®

If we choose $P_{\text{motor}} = 30 \text{ kW}$ ~~23.07~~ 23.07 kW

$$\begin{aligned} P_{\text{Rms load power}} &= 18.28 / \cancel{23.07} \times 100\% \\ &= 60.90\% \\ &= 79.02\% \end{aligned}$$

Choose a ~~60~~ 23.07 kW motor

23.07 kW .

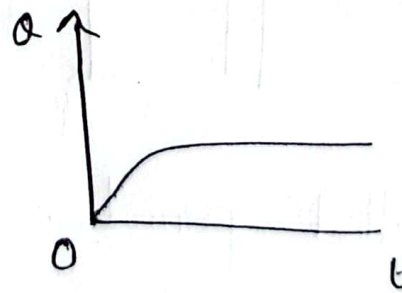
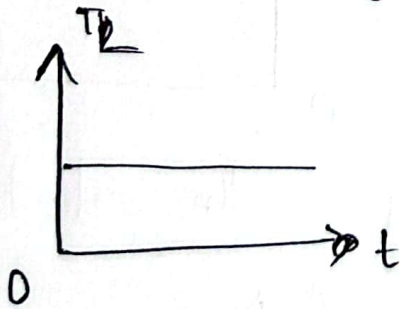
Moton duty class

Moton duty class defines the load cycles of the electric moton

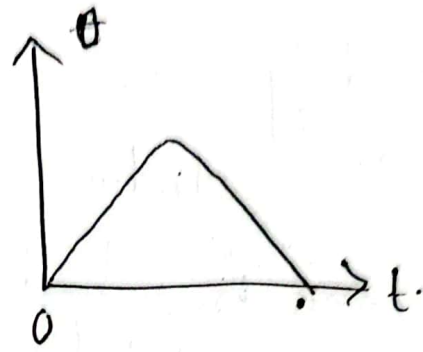
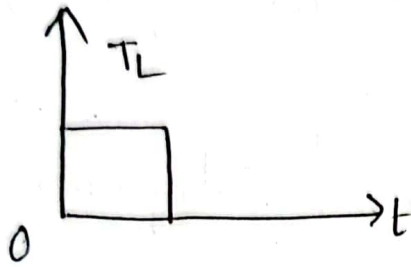
moton can be dividen in 8 Calagories :

- ① Continuous duty.
- ② ~~Stop~~ Short time duty.
- ③ Intermittent periodic duty.
- ④ Intermittent periodic duty with starting.
- ⑤ Intermittent periodic duty with starting & braking.
- ⑥ Continuous duty with intermittent periodic load.
- ⑦ Continuous duty with starting & braking.
- ⑧ Continuous duty with periodic speed change.

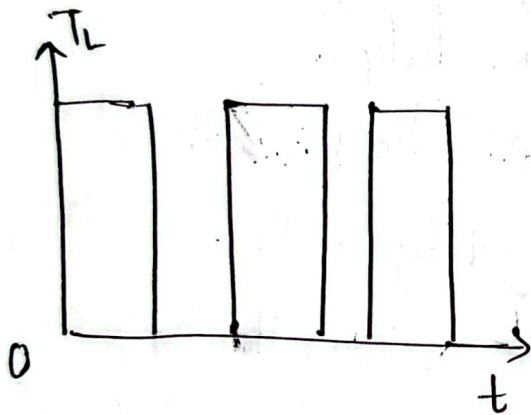
1) Continuous duty



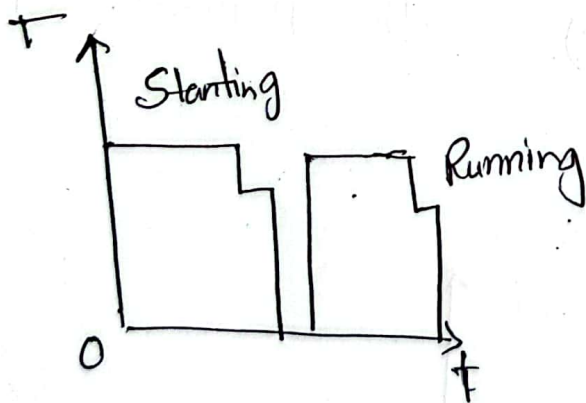
2) Short time duty.



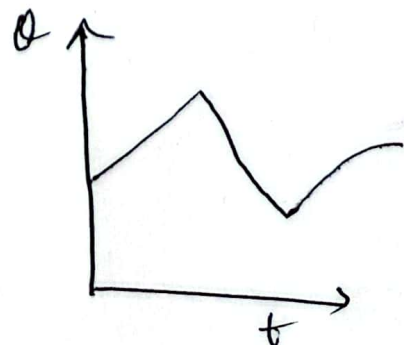
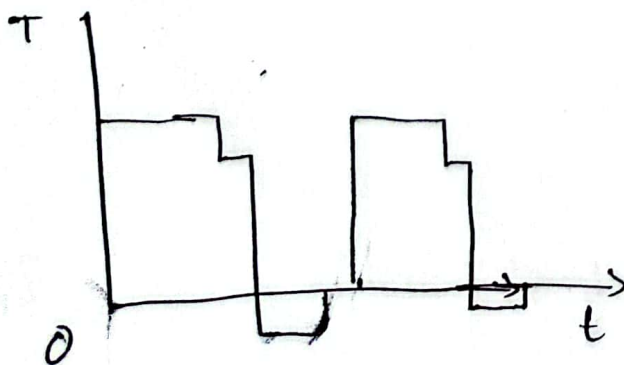
3) Intermittent periodic duty.



4) Intermittent periodic duty with Starting:



5) Intermittent periodic duty with Starting & brake.



Faraday's law

1st law: If a flux passes through a turn of a coil of wire, the voltage will be induced in the turn of the wire that is directly proportional to the rate of change in the flux with respect of time.

$$e_{ind} = - \frac{d\Phi}{dt}$$

2nd law: States that induced emf is equal to the rate of change on leakage on flux cut.

$$e_{in} = - \frac{d\Phi}{dt}$$

$$e_{ind} = - N \frac{d\Phi}{dt} \text{ where -}$$

e_{ind} = Induced voltage induced in the coil

N = Number of turn in coil

Φ = flux passing through coil.

Flemming left / Right hand Rule:

Thumb $\rightarrow$ Force

Index $\rightarrow$ Magnetic field

Middle finger $\rightarrow$ Current

Flemming left hand rule: It states if we place our thumb, forefinger and middle finger of the left hand perpendicular to each other, then the thumb points towards the direction of force by the conductor, the forefinger points towards magnetic field, and middle finger points towards direction of electric current.

Flemming Right hand Rule: It states that, if we place our thumb, forefinger and middle finger of the right hand perpendicular to each other, then the thumb points towards the motion of the conductor relative to the magnetic field, the middle forefinger points the direction of magnetic field and the middle finger points the direction of induced current.